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SECP2613-02 SYSTEM ANALYSIS AND DESIGN (WBL)

PROJECT 1 PHASE 2

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Table of Content

1.0 OVERVIEW OF THE PROJECT.....	3
2.0 PROBLEM STATEMENT.....	4
3.0 PROPOSED SOLUTIONS.....	5
3.2.2 M-CHAT-R Screening Integration.....	6
3.2.3 Automated Scoring Algorithm of Risk Level.....	6
3.3.1 Automated Notification System.....	7
3.3.2 Reporting and Analytics.....	7
3.3.3 Behavior Monitoring.....	7
4.0 INFORMATION GATHERING PROCESS.....	9
4.1 Method used.....	9
4.1.1 Interactive methods.....	9
4.2 Summary from method used.....	9
4.2.1 Documented Interview Transcript.....	9
4.2.2 Key Findings and Requirements Derivation.....	10
5.0 REQUIREMENT ANALYSIS.....	12
5.1 Current business process.....	12
5.1.1 Scenario.....	12
5.1.2 Workflow.....	13
5.2 Functional Requirement.....	14
5.2.1 Context Diagram.....	14
5.2.1 Diagram 0.....	14
5.2.2 Child Diagram: Process 1.0, Conduct M-CHAT-R Screening.....	15
5.3 Non-functional Requirement (NFR).....	16
5.3.1 Performance.....	16
5.3.2 Control.....	17
5.4 Logical DFD AS-IS system.....	18
6.0 SUMMARY OF REQUIREMENT ANALYSIS PROCESS.....	21

1.0 OVERVIEW OF THE PROJECT

It cannot be understated how vital early diagnosis is when it comes to providing timely assistance for children with autism. However, even though this aspect is extremely important, it should not be overlooked that many of the approaches that are currently being utilized in relation to the problem tend to be scattered, expensive, and unavailable both for families and healthcare workers alike.

The suggested idea for the project revolves around developing a prototype called "Autism Early Screening Support System". The purpose of creating such a system is to provide a technological means that could facilitate the screening process, make it easily accessible, and offer accurate information. In order to make sure that the system becomes widely available to potential users, it will be developed for both the mobile platform and the web.

The system will benefit three main stakeholder groups, which include parents, autistic children, and a non-profit organization (NGO). As for the system owner, the NGO will own the system. The core functionality of the system will involve the incorporation of different modules, such as M-CHAT-R (Modified Checklist for Autism in Toddlers, Revised) autism screening module and profiling according to the BPPOKU Pindaan 2019 (Bahagian A). This system will seek to solve the present challenges in data management, which are impeding efficient and systematic provision of services by both the health care and education sectors.

2.0 PROBLEM STATEMENT

The efficient support of autistic children becomes possible only due to the timely screening in the early stages of life. The main issues that arise when dealing with autism screening processes include the following:

2.1. The Disjointed and Unavailable Nature of the Process

Current methods of assessing whether a kid is developing normally require excessive resources and accessibility problems. Moreover, the fragmented nature of the process may make it difficult for parents to find out what to do first when suspecting that their child is underdeveloped.

2.2 Overreliance on Traditional Screening Practices

Despite the availability of more advanced technologies, traditional screening procedures are still very dependent on manual labor. Due to this fact, as well as other reasons related to the lack of communication and cooperation, screening often becomes delayed.

2.3 Existence of Systemic Obstacles and Absence of Integration

In most cases, there is a low degree of awareness as well as the absence of standardization. In addition, the lack of integration between the health care and education sectors leads to a "silo mentality" situation when the exchange of information by the parties accountable for a child's development is rather poor.

2.4 No Technology-Based Centralization Solution

The lack of centralization, with no technology-based solution, leaves room for inefficiencies in terms of data management as well as coordination of activities by various NGOs and parents. Tracking of results for screening tests, such as M-CHAT-R, and managing profiles based on BPPOKU guidelines will not be efficient without technological solutions.

As a result of such issues, the problem area described above is one wherein a delay in providing support for children with symptoms of autism may occur.

3.0 PROPOSED SOLUTIONS

The recommended solution is known as the Autism Early Screening Support System, which is an advanced two-fold application Web and Mobile technology system. The system combines standard medical screening procedures with daily management systems to create an efficient technological platform for the use of parents, children, and NGOs.

3.1 Core System Infrastructure and Security

The basis of the system lies in data security and privacy.

3.1.1 Authentication and Access Controls

The system uses safe methods for logging into and out of the system, which ensures that confidential data is protected by not allowing access to people who are not authorized to see the data.

3.1.2 Registration and Account Management

A separate module of the system is used to manage users' accounts.

3.1.3 Data Security and Privacy

Considering the confidentiality of data in question, data security is a critical feature of the core infrastructure of the proposed solution, as it will serve as a reliable environment for the formative years intervention program.

3.2 Standardized Screening & Profile Management

This module digitalizes the old screening processes manually performed to guarantee accuracy and fast intervention.

3.2.1 Digital Profile Management (BPPOKU Pindaan 2019)

The program will digitalize Bahagian A (Biodata) from the BPPOKU (Pindaan 2019) form. This process will include the following important pieces of information:

- Personal Details

Name, MyKad / MyKid identification, and contact details.

- Education & Professional Experience

Education level and parents/guardians' employment.

- Details of Next of Kin

The system collects detailed information about relatives close to the person for further communication in case of necessity.

3.2.2 M-CHAT-R Screening Integration

This feature includes a test with 20 questions performed according to the M-CHAT-R standard. The users provide "Yes" or "No" answers regarding children's behavior, such as do they react to a gesture or respond when called?

3.2.3 Automated Scoring Algorithm of Risk Level

The system will be able to provide risk levels automatically, making sure that each child undergoes an evaluation process that is not only consistent but also standardized and objective in nature. For those children with results falling within the Low Risk category, which would include any score of 0-2, there would not be a need to pursue any additional actions unless there are any future worries. Those with scores ranging from 3-7 are considered to have Moderate Risk, automatically prompting notification about the requirement of taking some actions to further examine their behavior. In instances where the scores range from 8-20, the result will be classified as a High Risk automatically, skipping other steps to indicate the urgent requirement of referral of the client for professional diagnostic testing.

3.3 Support, Analytics and Specialized Management

In addition to the diagnosis, this system features modules which enable monitoring and assistance on a continuous basis.

3.3.1 Automated Notification System

To resolve the issue of "delayed referrals" in the conventional approach, this module helps generate automated alerts regarding upcoming screenings or follow-up consultations.

3.3.2 Reporting and Analytics

Data analytics provide an opportunity to understand how the development of a child evolves throughout time, as visualized by parents or NGOs.

3.3.3 Behavior Monitoring

This module monitors the emotional state and mood of the child, enabling caregivers to identify any possible triggers for certain behaviors. These are some of the functions of this module:

- Behavior Log Processor

A real-time function that enables the caregiver to log emotional status, particular behavioral instances, and the corresponding environmental antecedents.

- Trigger Analyzer

A pattern-matching algorithm that scans historical logs to identify correlations between environmental factors that repeat themselves and behavioral disruptions that occur again.

- Mood Trend Evaluator

A computation function that calculates and charts baseline emotional movements over specified timelines, providing parents and NGOs with empirical evidence of structural improvement.

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3.3.1 Appointment Scheduling and Referral Management

This module allows parents to book appointments directly with specialist and referral centres, as well as with education centres, through the system. When a child's behavioural log or screening result shows a concerning trend, parents can make an appointment without needing to rely on a phone call, resolving the ambiguity and delay parents currently face once the limited monthly clinical sessions come to an end. The module checks real-time availability of the specialist or referral centre, records the appointment slot, and automatically issues a referral letter and appointment reminder to the relevant party. This directly replaces the current appointment method and the phone call and manual diary check process, which currently takes 1 to 3 working days and produces no automatic reminder.

4.0 INFORMATION GATHERING PROCESS

4.1 Method used

4.1.1 Interactive methods

In order to gain in-depth qualitative understanding of the current operational model and user constraints, the team conducted structured interviews as the core interactive information-gathering method. It is chosen as an opportunity to establish a qualitative dialogue with those parents and guardians who are involved in monitoring early childhood development. As a result of such a direct contact with the main stakeholder, the team has managed to gain in-depth and unmediated information about obstacles, frictions, and fears associated with the use of the traditional paper-based method of diagnosing autism. The qualitative data collected in the course of such interviews helped outline important parameters, namely the lack of understanding of manual scoring, communication barriers during the waiting period for feedback, and expectations regarding the user experience.

4.2 Summary from method used

4.2.1 Documented Interview Transcript

Interviewer: Good afternoon, Madam Ling. Thank you for joining me today. My name is Ng Xuan Yee, a student from Universiti Teknologi Malaysia. Our team is researching how families track early childhood development markers and manage home monitoring. To begin, could you introduce yourself and share a bit about your children?

Interviewee: Good afternoon. I am a housewife taking care of my two kids. My eldest child is already diagnosed with mild autism, while my younger daughter is currently displaying moderate development delays and behavioral traits. We are constantly monitoring both of them at home.

Interviewer: Under the current manual approach, what are the main challenges you face when tracking your children's behavioral development?

Interviewee: The major challenge is that of ambiguity. When a non-verbal child throws tantrums or gestures towards objects, it leaves the parents in a situation where they have to guess all the time. Without an organized log, it becomes extremely hard to distinguish between normal toddler tantrums and signs of autism development.

Interviewer: How have you been managing their communication and behavioral needs up to this point?

Interviewee: I use do-it-yourself visual aids manually. I make flashcards and hand signals for associating words with the actions they want to do. It is really tiring keeping track of all these paper cards in my home.

Interviewer: Are your children currently receiving regular clinical therapy or hospital appointments?

Interviewee: No, the hospital therapy sessions have concluded, so continuous monitoring falls entirely on us. In previous months, all we had was a single-hour-long clinical session once a month. Once those sessions end, there will be no progress tracking methods available to parents.

Interviewer: Based on these challenges, what specific features and design style do you expect to see inside our proposed mobile application?

Interviewee: The system must contain a visual behavioral diary. First, the system must have colorful and interactive buttons representing various behaviors in the child's day-to-day life. Second, it is absolutely necessary for us to have a fully integrated booking system in this center. In case our behavioral log shows some trends which alarm us, we need to be able to make an easy appointment in order to talk to the professional organizations regarding our child. The design of such a system has to be visually attractive but not complicated in order to allow parents and children to use it easily.

Interviewer: This has been incredibly insightful, Madam Ling. Your input gives our team concrete data to map out our functional requirements. Thank you so much for your cooperation.

Interviewee: You are very welcome. I hope this system can truly help families streamline home monitoring.

4.2.2 Key Findings and Requirements Derivation

From the interview carried out with the participant, who is a mother to two children with development variations, some important aspects have been found concerning behavioral logging, communication problems during the day, and access to expert professionals.

From the analysis of the current manual system, it is evident that there is significant confusion among the parents as far as assessment of the child development is concerned since there is a need to distinguish between a normal temper tantrum and autism based on parent guesses and no structured log to refer to. In addition, using homemade flash cards and manual hand signs and physical aids to help overcome the communication barrier between the children is extremely physically and mentally demanding for caregivers. It becomes even worse after the completion of the clinically guided therapy which lasts only for one hour per month. It is because when the short month-long sessions are over, the families are stranded in a vacuum of isolation without any way to track their progress or set up an appointment to seek advice from the center professionals.

In order to address these gaps in operations, the interview process helped in identifying system specifications which actually lay out the business rules for the system. In terms of inputs, the system should have a visually oriented simple user interface with colorful and

interactive buttons which enable the parents to input their daily behavioral data in an online diary and also a center appointment booking system that enables parents to schedule their desired consultation timings and organization preferences. The processing involves a backend analytics program that translates daily input data into visual scannable progress reports and is done in parallel with a scheduling program that enables real-time slot allocation for partner organizations. The outputs will include simplified visual behavioral trends for the parents' dashboard in order to help them track trends without any difficulty and digital appointment booking confirmations that will make it easy for parents to book their appointments with centers when alarming trends are found.

5.0 REQUIREMENT ANALYSIS

5.1 Current business process

5.1.1 Scenario

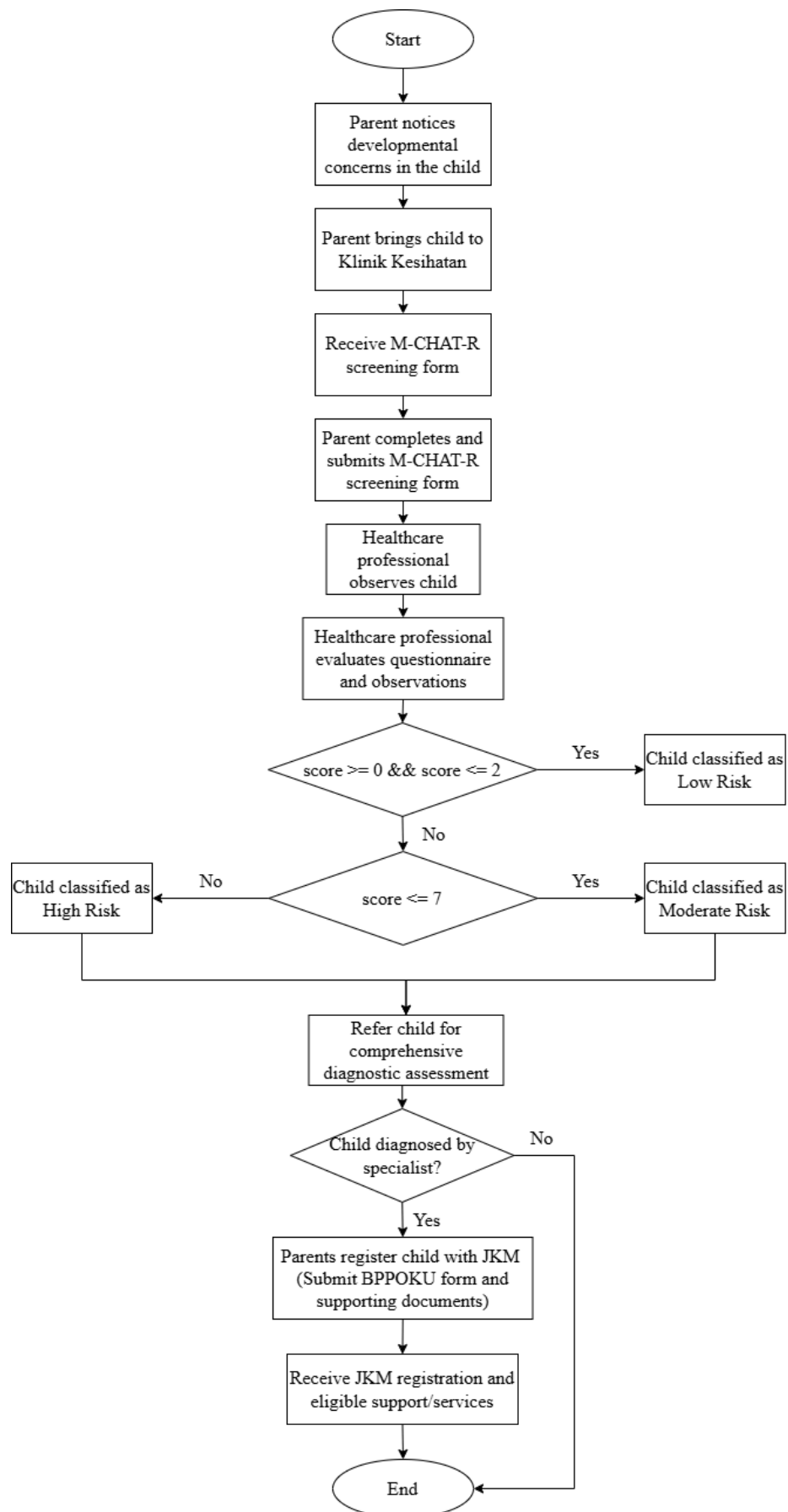
Nowadays, the current system relies heavily on manual processes and traditional methods. The current autism early screening support process is primarily conducted through healthcare professionals, educators or autism specialists. Parents or guardians who suspect developmental delays in their children are required to visit a healthcare centre, clinic or autism intervention centre for an initial assessment.

During the screening session, parents are required to complete paper-based questionnaires which are known as M-CHAT-R regarding the child's communication skills, social interaction, repetitive behaviours, and developmental milestones. Healthcare professionals then observe the child's behaviour and review the completed questionnaire together with the parents.

Based on the collected information, the healthcare professional performs a preliminary screening to determine whether the child shows signs associated with Autism Spectrum Disorder (ASD). If the screening indicates possible autism traits, the child is referred for a comprehensive diagnostic assessment conducted by qualified specialists. If the child is formally diagnosed with Autism Spectrum Disorder (ASD) by a qualified specialist, the parents or guardians may register the child with the Department of Social Welfare (Jabatan Kebajikan Masyarakat, JKM). To complete the registration, they are required to submit the BPPOKU form together with the necessary supporting documents, such as the medical diagnosis report and identification documents. Upon successful registration, the child may become eligible for welfare assistance, support services, and benefits provided by JKM.

The current process has several limitations. The screening process is time-consuming, relies heavily on manual documentation, and requires parents to attend physical appointments. In addition, there is limited accessibility to specialists in certain areas, and parents often experience delays before receiving screening results or recommendations for further assessment.

5.1.2 Workflow



5.2 Functional Requirement

5.2.1 Context Diagram

Process	Input	Output
Manual Autism Screening & Record Keeping System	- Child Behavioural Responses - Children details - Completed M-CHAT-R - Children Behavioural Patterns - Filled BPPOKU (BHG A) form - Scored Checklist - M-CHAT-R Checklist	- Screening Result - Risk Level Classification - Notification Reminders - Profile & Behaviour Records - Manual Referral Letter - Feedback & Diagnostic Results - Compiled Records

5.2.1 Diagram 0

Process	Input	Output
1.0 Complete M-CHAT-R Screening	- Child behavioural responses - Completed M-CHAT-R - Past Screening Records - Scored Checklist Results	- Risk level result - Printed checklist & risk classification - Completed Checklist - Screening Tests - Screening Summary
2.0 Manage Child Biodata Profile	- Biodata (BPPOKU) Details - Request to retrieve existing profile - Parent/caregiver biodata submission	- Stored Biodata Profile - Updated child profile - Biodata Summary
3.0 Record Behaviour Observations	- Behavioural incidents observed from child	- Stored Behaviour Notes - Behaviour Summary

	- Past observation notes - Parent/caregiver notes for behavioural incidents	
4.0 Schedule Appointments and Referrals	- Referral instruction - Availability of specialist/referral center - Diagnostic Feedback	- Referral Letter - Appointment Reminder - Appointment Summary - Record appointment slot
5.0 Compile Reports	- Screening Summary - Behaviour Summary - Biodata Summary - Appointment Summary - Report Request	- Compiled Reports

5.2.2 Child Diagram: Process 1.0, Conduct M-CHAT-R Screening

Process	Input	Output
1.1 Distribute Screening Checklist	- Provided Checklist Template	- Blank Checklist
1.2 Complete 20 questions Checklist	- Parent's Yes/No Answers - Observed Behavioural Response	- Screening Test - Completed Checklist
1.3 Manually Score Checklist	- Completed Checklist	- Total Score Tally
1.4 Determine Risk Level	- Total Score Tally	- Risk Level classified
1.5 Result & Follow-Ups	- Risk Level classified	- Manual Referral Letter - Result & score sheet - Result & Recommended Action Steps

		- Stored completed & scored checklist
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5.3 Non-functional Requirement (NFR)

The non-functional requirements below describe the performance and control characteristics observed in the current (AS-IS) manual autism screening and record keeping process, prior to the introduction of the newly proposed Autism Early Screening Support System. Each requirement is written as a specific statement that defines the system properties and constraints and are derived from the team's background study along with interviews and observation findings on how parents, healthcare workers, and the NGO are currently carrying out the screening, profiling, and behaviour tracking using paper based methods.

5.3.1 Performance

Type of NFR	Requirement Statement	Metric
Product	Manually tallying and classifying one completed M-CHAT-R checklist shall take the healthcare workers approximately 5-10 minutes per child to tally and classify a completed checklist by hand.	Speed
Product	The physical filing system shall require staff to manually search paper records, taking from several minutes up to over an hour to retrieve a child's past screening or profile record, depending on the file volume.	Speed
Organizational	The manual record relaying process shall rely on phone calls or physical document delivery to exchange a screening result or biodata record between the clinic or NGO and an external party, taking 1 to 3 working days to complete.	Speed
Product	The manual behaviour tracking process shall require a caregiver or NGO staff member to review multiple weeks of handwritten notebook entries by hand to identify a behavioural pattern, with no automated trend computation provided.	Speed/ Ease of Use
Product	The manual appointment scheduling process shall rely on a phone call and a manual diary check to confirm an appointment slot or issue a reminder, with no automatic reminder generated.	Speed
Product	The M-CHAT-R checklist and BPPOKU form shall be presented only in a fixed paper format with no guided or interactive assistance, requiring staff support for parents	Ease of Use

	with limited literacy or language barriers.	
Organizational	The manual screening and record keeping process shall require proportionally more staff time and physical storage as the number of registered children increases, with no mechanism provided to absorb growth in volume.	Scalability

5.3.2 Control

Type of NFR	Requirement Statement	Metric
Organizational	The physical filing cabinet shall store a child's confidential MyKad number and family details without any authentication or role based access restriction, allowing any staff member with physical access to view the record.	Reliability
Organizational	The manual record keeping process shall not maintain an audit log of who accessed, copied, or amended a paper record, as handwritten annotations cannot be traced to a specific user or date.	Reliability
Product	The manual M-CHAT-R scoring process shall rely on manual tallying with no built-in validation, leaving the score classification subjected to human error.	Robustness
Organizational	The paper based record system shall retain only a single physical copy of each record with no backup, resulting in permanent loss of that record if it is lost, damaged, or misfiled.	Robustness
Organizational	The manual filing process shall not enforce a standard completeness check on the BPPOKU Bahagian A form, allowing forms with incomplete fields to be filed.	Robustness
Organizational	The manual scoring process shall depend on individual healthcare workers to score and classify checklists by hand, with no standardised, enforced rule to ensure consistent classification across staff members.	Consistency

5.4 Logical DFD AS-IS system

The Data Flow Diagram (DFD) below models the current (AS-IS) manual workflow of the autism screening and record keeping process, based on the business process described in 5.1.

5.4.1 Context Diagram (Level 0)

The Context Diagram shows the AS-IS “Manual Autism Screening & Record Keeping Process” as a one process that depicts the entire system, together with four external entities to exchange information with, which are the Parents/ Caregivers, the Healthcare Worker, the NGO Administrator, and the Specialist/ Referral Center. The Child entity is also included as the healthcare worker directly observes the child’s behavioural patterns during the screening interaction, in addition to information supplied by the Parent entity.

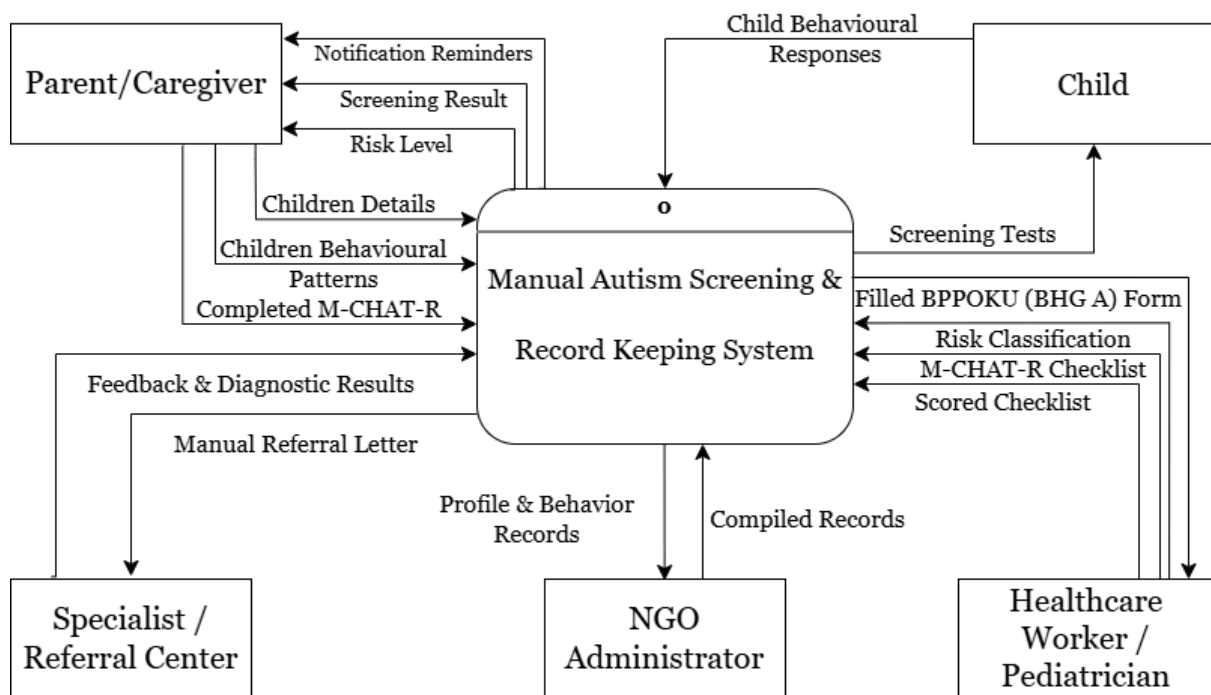


Figure 5.1 Context Diagram (Level 0) AS-IS Manual Autism Screening & Record Keeping System

5.4.2 Diagram 0

The Diagram 0 expands the single AS-IS process into five major manual process and four manual data stores :

- 1.0 Conduct M-CHAT-R Screening: Distribute, complete and manually score the 20 item checklist.
- 2.0 Manage Child Biodata Profile: Collecting and filling the BPPOKU (Bhg A) biodata form
- 3.0 Record Behaviour Observations: Parents manually record behavioural incidents
- 4.0 Schedule Appointments and Referrals: Booking follow-up visits and issuing referrals
- 5.0 Compile Reports: NGO Staff manually compiling screening, behaviour, and appointment records into a report.

The four data stores involved are the D1 Screening Form Files, D2 Child Profile (BPPOKU) Files, D3 Behaviour Log Notebooks, and D4 Appointment Record Book.

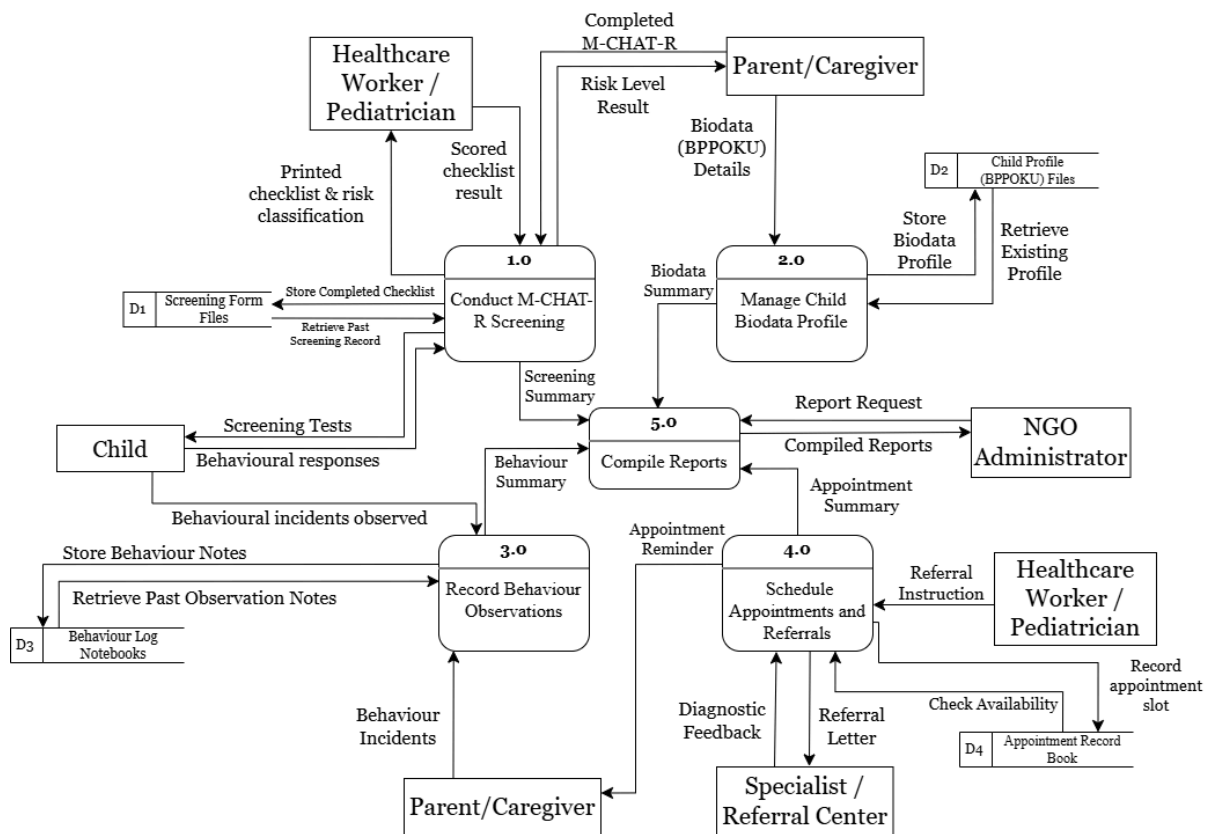


Figure 5.2 Diagram 0 AS-IS Manual Autism Screening & Record Keeping System

5.4.3 Child Diagram: Process 1.0, Conduct M-CHAT-R Screening

Process 1.0 was further expanded into a child diagram as it is the most detailed and critical process in the current manual workflow that involves the 20 question M-CHAT-R checklist, manual scoring, and the risk classification level:

- 1.1 Distribute Screening Checklist: Healthcare workers distribute the checklist form to the parents.
- 1.2 Complete the 20 question Checklist: Parents answer Yes/ No for each of the 20 M-CHAT-R questions based on the child's observed behavioural responses.
- 1.3 Manually Score Checklist: Healthcare Worker tallies the Yes/ No responses by hand.
- 1.4 Determine Risk Level: Healthcare worker classifies the score into Low/ Moderate/ High risk.
- 1.5 Result and Follow-Ups: Healthcare workers inform the parents and issue a manual referral letter to a specialist for High Risk cases.

The four data stores involved are the D1 Screening Form Files, D2 Child Profile (BPPOKU) Files, D3 Behaviour Log Notebooks, and D4 Appointment Record Book.

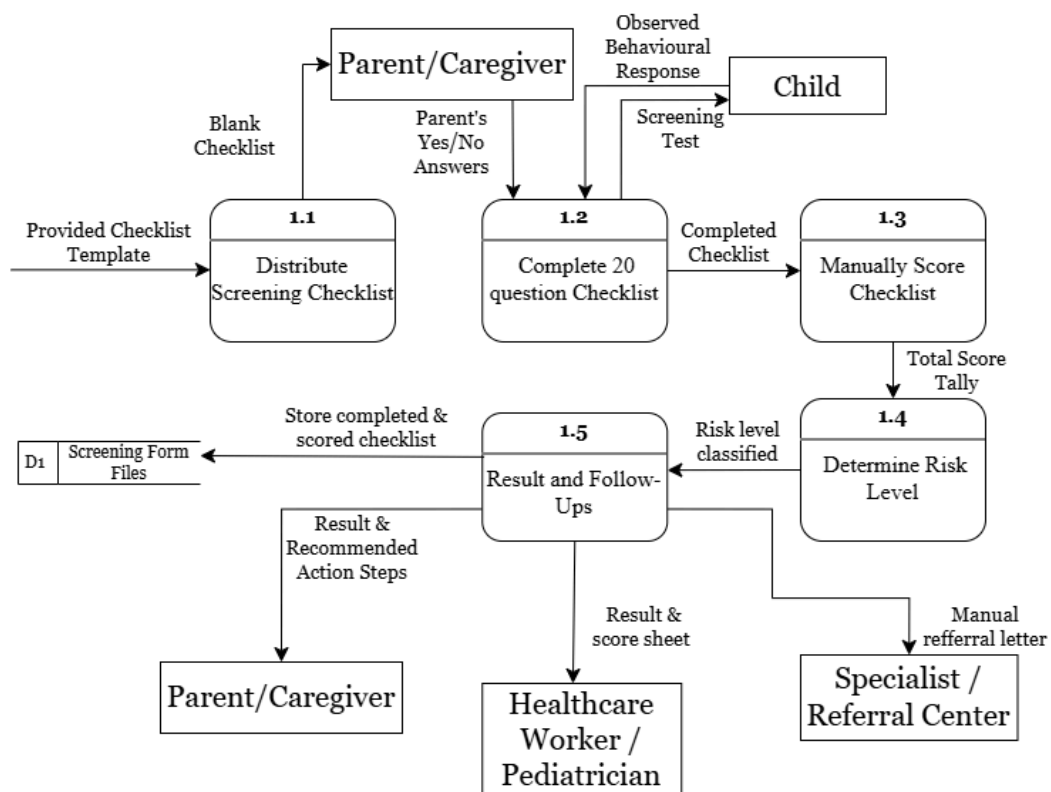


Figure 5.3 Child Diagram (Process 1.0) AS-IS Manual Autism Screening & Record Keeping System

6.0 SUMMARY OF REQUIREMENT ANALYSIS PROCESS

In Phase 2, this project focused on gathering and analysing information about the current (AS-IS) autism screening and record keeping process before any new system is introduced. By conducting an interactive method, in the form of a structured interview with a parent of two children with developmental variations, several issues were discovered, where parents struggle to distinguish typical behaviour from early signs of autism without a structured log, and communication support relies entirely on tiring, homemade physical aids and families are left without any tracking or referral mechanism once the monthly clinical sessions conclude.

These findings were translated into a documented AS-IS business process, functional requirements expressed as process, input, and output tables for the Context Diagram, Diagram 0, and the Child Diagram, and a set of non-functional requirements covering the performance and control characteristics of the manual process. The Logical DFD further confirmed that the current process depends on five manual activities and four paper-based data stores, none of which offer validation, backup, or role based access control.

Overall, the requirement analysis process shows that the current manual system is functional but slow, error-prone, and difficult to scale, and that its main weaknesses lie in absence of any follow-up mechanism once formal clinical sessions conclude as well as lack of automation, security, and centralization. These AS-IS findings, together with the proposed solutions outlined, will form the basis for designing the TO-BE Autism Early Screening Support System in the next phase of the project.